WSAVA
WORLD SMALL ANIMAL VETERINARY ASSOCIATION

Core and Non-Core Vaccines for Pet Dogs (non-shelter) according to the 2024 guidelines for the vaccination of dogs and cats – compiled by the World Small Animal Veterinary Association (WSAVA) Vaccination Committee.

CORE VACCINES FOR PET DOGS

Vaccine	Puppies ≤ 16 weeks	Dogs > 16 weeks	Revaccination	Comments and Recommendations
• Canine Parvovirus-2 (CPV; modified live virus - MLV) + • Canine Distemper Virus (CDV; MLV or recombinant) + • Canine Adenovirus-2 (CAV-2; MLV): parenteral	Start no earlier than 6 weeks of age. Revaccinate every 3–4 weeks until 16 weeks of age. In especially high-risk situations continue until 20 weeks of age and consider vaccinating every 2–3 weeks.	Two doses 2–4 weeks apart are recommended by some manufacturers. However, a single dose of MLV or recombinant vaccine will likely protect most dogs.	Consider revaccinating at about 6 months of age, rather than waiting until the dog is 12-16 months of age. This will narrow the window of susceptibility for any puppies that failed to mount an active immune response earlier. Thereafter, revaccinate at 3 years of age and thereafter no more frequently than every 3 years.	These core vaccines are among the most important received by puppies and dogs. The aim should be to vaccinate as large as possible a proportion of the entire population. Careful socialisation of puppies (during their sensitive period for socialisation) can begin before the completion of this vaccination series.
• Canine Parvovirus-2 (CPV; recombinant) + • Canine Distemper Virus (CDV; MLV): parenteral	Administer a single dose from 4 weeks of age prior to commencing routine primary vaccinations.	-	-	This recently introduced product is aimed specifically at young puppies, likely to have interfering MDA, rather than for revaccination of older dogs.
• Rabies (inactivated): parenteral	Follow any local laws or regulations as a priority. Follow the product leaflets of locally manufactured vaccines. In some countries, the first dose is generally not given before 12 weeks of age.	Follow any local laws or regulations as a priority. Follow the product leaflets of locally manufactured vaccines.	Follow any local laws or regulations as a priority. Follow the product leaflets of locally manufactured vaccines. Revaccination at 1 year of age (or in some countries 1 year after the primary vaccination) is required. Canine rabies vaccines with either a 1- or 3-year DOI are available. Timing of boosters is determined by the licensed DOI, but in some areas may be dictated by law.	Core wherever the disease is endemic or wherever local laws or regulations require.
• *Leptospira* spp. (killed bacterin): parenteral • The serogroups included in vaccines depend on the geographical region. Most vaccines include at least 2 serogroups, but some are monovalent, some trivalent and some quadrivalent.	Initial dose is usually from 8 weeks of age. Follow the advice in the product leaflet about when to start. A second dose is given 2–4 weeks later.	Two doses 2–4 weeks apart.	Annually.	Core for dogs in regions where canine leptospirosis is endemic, implicated serogroups are known, and suitable vaccines that include implicated serovars are commercially available.

NON-CORE VACCINES FOR PET DOGS

Vaccine	Puppies ≤ 16 weeks	Dogs > 16 weeks	Revaccination	Comments and Recommendations
• Canine Parainfluenza Virus (CPiV; MLV): parenteral	Administer from 6 weeks of age onwards, then every 2–4 weeks until 16 weeks of age or older.	Two doses 2–4 weeks apart are generally recommended by manufacturers.	Annually.	The duration of immunity provided in pet dogs is uncertain. Use of CPiV (MLV-mucosal) in combination with *Bordetella bronchiseptica* may be preferable as CPiV is not broadly available as a single antigen product.
• *Bordetella bronchiseptica* (live avirulent bacteria): intranasal • *B. bronchiseptica* (live avirulent bacteria) + Canine Parainfluenza Virus (CPiV; MLV): intranasal • *B. bronchiseptica* (live avirulent bacteria) + Canine Parainfluenza Virus (CPiV; MLV) + Canine Adenovirus-2 (CAV-2; MLV): intranasal • *B. bronchiseptica* (live avirulent bacteria): oral • *B. bronchiseptica* + Canine Parainfluenza Virus (CPiV; MLV): oral	Each of these mucosally applied vaccines provides protection after a single dose. Some can be applied as early as 3 weeks of age, others at 7 or 8 weeks of age. Follow the product leaflet advice.	A single dose.	Annually.	These live vaccines intended for intranasal or oral administration MUST NOT be inadvertently injected parenterally as this may lead to a severe adverse reaction.
• *Bordetella bronchiseptica* (killed bacterin): parenteral • *Bordetella bronchiseptica* (cell wall antigen extract): parenteral • *Bordetella bronchiseptica* (fimbrial antigen): parenteral	These parenteral bacterin or subunit vaccines require two doses to immunize, generally given 2 to 4 weeks apart. Review and follow product leaflet intervals.	Two doses. Review and follow the specific product leaflet intervals as there are differences between products.	Annually for most. Review and follow the specific product leaflet intervals. There are special revaccination instructions for the fimbrial antigen vaccine.	These vaccines intended for parenteral use should not inadvertently be given intranasally or orally. This would not be effective and might cause unnecessary discomfort.
• *Borrelia burgdorferi* (Lyme borreliosis; killed whole bacterin: parenteral • *Borrelia burgdorferi* (subunit-Outer surface protein A - OspA): parenteral • *Borrelia burgdorferi* (subunit OspA and chimaeric OspC proteins): parenteral	These parenteral bacterin or subunit vaccines require two doses to immunize, generally starting at about 8 weeks of age. Review and follow the specific product leaflet intervals.	Two doses 2-4 weeks apart. Review and follow the specific product leaflet intervals.	Annually. Revaccinate just prior to the start of the tick season, as determined regionally.	Generally recommended only for use in dogs with a known high risk of exposure, living in or visiting regions where the risk of vector tick exposure is considered to be high, or where disease is known to be endemic. The mainstay of Borrelia prevention is diligent ectoparasite control. Direct dog to dog transmission does not occur.
• Canine influenza virus (H3N8; killed adjuvanted): parenteral • Canine influenza virus (H3N2; killed adjuvanted): parenteral • Canine influenza virus (Bivalent H3N8 + H3N2; killed, adjuvanted): parenteral	These parenteral, inactivated viral vaccines require 2 doses to immunize. The initial dose can be given from 6 weeks of age. Review and follow the specific product leaflet intervals.	Two doses. Review and follow the specific product leaflet intervals, generally 2–4 weeks apart.	Annually.	Licensed only in USA. Consider for use in at-risk groups of co-housed dogs such as those in kennels, dog shows or day care.
• Canine leishmaniosis (CanL; recombinant protein A2): parenteral • CanL (excreted-secreted proteins -LiESP- of *L. infantum*): parenteral • CanL (recombinant protein Q): parenteral	• Three doses, 3 weeks apart with the initial dose at >4 months of age. • Three doses, 3 weeks apart with the initial dose at >6 months of age. • A single dose at >6 months of age.	• Three doses, 3 weeks apart. • Three doses, 3 weeks apart. • A single dose.	Annually.	Prevention of canine leishmaniosis depends crucially on diligent ectoparasite control so as to minimise contact with the vectors. Vaccination can be regarded as a supplementary control measure, not a substitute for diligent ectoparasite control.
• Canine herpesvirus-1 (CHV-1; subunit): parenteral	Not applicable. This vaccine is intended for pregnant bitches.	Two doses during pregnancy. First dose – During oestrus or 7-10 days after the presumed date of mating. Second dose – 1 to 2 weeks before the expected date of whelping. Follow the product leaflet.	Manufacturer recommends repeating the 2-injection protocol during each subsequent pregnancy.	This vaccine is intended to protect newborn puppies. Infection (usually from the dam) can be fatal in young puppies, <3 weeks old, that are not kept warm.

Vaccines against canine enteric coronavirus, *Giardia* spp and *Microsporum canis*, available in some countries, are considered not recommended for pet dogs.